

Algebra 2

Slope, Revenue & Area Drill — Functions & Logarithms

Name: _____

Block: _____

Date: _____

~35 minutes • 46 points total. Calculator allowed on Q2. Show all work. Logarithm problems (Q4–Q7) are no calculator — rewrite both sides with a common base.

Q1 — Interpreting Slope on a Mountain Road

8 pts

Percent grade = (vertical drop $\div$ horizontal distance) $\times$ 100%. Show your work.

1.

A mountain road drops **42 feet** in elevation for every **700 feet** of horizontal distance.

(a) What is the percent grade of this road? Show your calculation.

Grade = _____ %

(b) Write a linear equation for vertical drop V (feet) as a function of horizontal distance H (feet).

$V =$ _____

(c) A driver starts at the top. After driving some distance, the altimeter shows the car is **84 feet lower** than the start. How far has the car traveled horizontally?

$H =$ _____ feet

(d) The base of the hill is **2,500 feet** horizontally from the top. What is the total height of the hill?

Height = _____ feet

Q2 — Revenue as a Function (Calculator)

12 pts

Expand, find zeros, find vertex, and use the quadratic formula. Show all work.

2.

A streaming service charges $(150 - 7x)$ dollars per subscription, where x is the number of subscriptions sold (in thousands). Revenue is price times quantity, so $R(x) = x(150 - 7x)$, measured in thousands of dollars.

(a) Expand $R(x)$ and write it in standard form $ax^2 + bx + c$. Identify a , b , and c .

$R(x) =$ _____

$a =$ _____ $b =$ _____ $c =$ _____

(b) What values of x make $R(x) = 0$? What do those values mean in context?

$x =$ _____

Meaning: _____

(c) Use the vertex formula $x = -\frac{b}{2a}$ to find the subscription count that **maximizes revenue**. Leave your answer as a fraction.

$x =$ _____ thousand subscriptions

(d) Calculate R at that maximum. Round to the nearest whole number (in thousands of dollars). What is the maximum revenue?

Max $R \approx$ _____ thousand dollars

(e) The company needs $R(x) = 700$ (thousand dollars). Set up the equation and solve using the quadratic formula. There are two solutions — what does each one mean for the business?

$x_1 \approx$ _____ thousand $x_2 \approx$ _____ thousand

Meaning: _____

Q3 — Triangle Under a Line

14 pts

Build the area function, explore it with a table, then find the maximum and solve for specific areas.

3.

The line $f(x) = 48 - 6x$ is drawn on a coordinate plane. A right triangle is formed with its base along the x -axis from 0 to x , and its height going straight up from $(x, 0)$ to the point $(x, f(x))$ on the line.

(a) Write the area function $A(x)$ using $A = \frac{1}{2} \cdot \text{base} \cdot \text{height}$. Simplify completely.

$A(x) =$ _____

(b) What is the domain of $A(x)$? Explain why x cannot go higher than its upper bound.

Domain: _____

(c) Complete the table.

x	1	2	3	4	5	6	7
$A(x)$							

What pattern do you notice in the table?

(d) Use the vertex formula to find the value of x that maximizes the area. What is the maximum area?

$x =$ _____ Max area = _____ square units

(e) Find all values of x where $A(x) = 45$. Factor to solve — no quadratic formula needed. Describe the two triangles.

$x =$ _____ and _____

(f) Find all values of x where $A(x) = 36$. For each solution, write the base and height of the triangle and explain why the two triangles have the same area even though they look different.

$x =$ _____ and _____

Explanation: _____

Q4 — Evaluate $\log_{27}(9)$ (No Calculator)

3 pts

Rewrite both 27 and 9 as powers of a common base, then set exponents equal.

4. Evaluate: $\log_{27}(9)$

$\log_{27}(9) =$ _____

Q5 — Evaluate $\log_{32}(8)$ (No Calculator)

3 pts

32 and 8 share a common base. Find it, rewrite both, then solve for the exponent.

5. Evaluate: $\log_{32}(8)$

$\log_{32}(8) =$ _____

Q6 — Evaluate $\log_{16}(32)$ (No Calculator)

3 pts

16 and 32 are both powers of 2. Write each that way, then set exponents equal.

6. Evaluate: $\log_{16}(32)$

$\log_{16}(32) =$ _____

Q7 — Evaluate $\log_{125}(25)$ (No Calculator)

3 pts

*125 and 25 share a common base — it's not 2 this time.***7. Evaluate:** $\log_{125}(25)$ $\log_{125}(25) =$ _____